



Hybrid Neural Network Based Signal Detection for Multi-User MIMO with Hardware Imperfections

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Hybrid Neural Network Based Signal Detection for Multi-User MIMO with Hardware Imperfections

Dawei Gao, Wenfeng Zhang, Qinghua Guo, Guisheng Liao, and Ming Jin

Abstract—This work deals with the issue of communication signal detection in multi-user multi-input multi-output (MIMO) with hardware imperfections, where low-cost users suffer from power amplifier nonlinearity (PAN) and in-phase and quadrature imbalance (IQI). To deal with the multi-user interference and nonlinear distortions due to hardware imperfections, we propose a novel signal detector using a hybrid neural network (HNN), which combines a Kolmogorov-Arnold network (KAN) and a conventional linear NN layer. The training of the HNN involves the update of both linear NN weights and nonlinear activation functions. It turns out that the proposed HNN detector with a much smaller model size significantly outperforms conventional polynomial-based techniques and state-of-the-art pure NN or KAN techniques.

Index Terms—Hardware imperfections, Kolmogorov-Arnold networks, multiple-input multiple-output, nonlinear distortion.

I. INTRODUCTION

AS a core technology of modern wireless communication systems, multiple-input multiple-output (MIMO) significantly enhances channel capacity and transmission reliability by effectively utilizing the spatial dimension. However, as communication systems evolve toward higher frequency bands, larger bandwidths, and more complex modulation schemes, hardware impairments in low-cost radio frequency (RF) front-ends have become increasingly prominent, posing a critical bottleneck to further performance improvement in MIMO systems [1], [2], [3]. Hardware impairments in low-cost RF front-ends significantly degrade the performance of MIMO systems, especially in higher frequency bands and complex modulation schemes [4].

Among various hardware impairments, in-phase/quadrature (I/Q) imbalance is one of the most common types, manifesting as mismatches between the I and Q signal paths, thereby disrupting the orthogonality of the transmitted signals [5]. Meanwhile, the nonlinear characteristics of power amplifiers (PAs) leading to severe signal distortion cannot be overlooked, which is particularly pronounced in millimeter-wave and terahertz communication systems [3]. The combined effects of these hardware impairments further exacerbate signal distortion, posing significant challenges to the receiver design of

MIMO systems. To address the distortion issues caused by hardware impairments, various solutions have been proposed. For the nonlinear distortion caused by the PA, some classic polynomials-based methods have been investigated [6], [7]. These methods approximate the nonlinear characteristics of power amplifiers (PAs) by constructing polynomial models, and apply either predistortion at the transmitter to compensate for the nonlinear effects introduced by PAs, or postdistortion at the receiver to mitigate the distortion [8]. However, these approaches are prone to the numerical instability issue due to the matrix inversion in the determination of polynomial coefficients [9]. Other methods using the Wiener model [10] and the Hammerstein model [11] have also been studied, however, these approaches often result in model mismatch issues. Moreover, most existing works address I/Q imbalance (IQI) or PA nonlinearity (PAN) individually [12], [13], despite their concurrent occurrence in many cases, which necessitates handling their combined effects.

Methods based on artificial neural networks (NNs) have demonstrated significant potential in addressing nonlinear distortion in communication systems, thanks to their universal approximation capabilities. In [14], a parallel multi-task learning (MTL) estimator is developed to mitigate the coupling effects among different hardware impairments by utilizing their mutual correlations, which proves effective in compensating IQI. The work in [15] first introduced a real-valued focused time-delay neural network (RVFTDNN) to simulate PA's behavior, achieving promising performance. Building on this, extensive research [16], [17] has explored enhanced versions of RVFTDNN. Among these, the augmented RVFTDNN (ARVTDNN) model [16] improves the mitigation of PAN and IQI by incorporating higher-order signal components into the input of RVFTDNN. However, an issue with the ARVTDNN based techniques is that a large number of pilot symbols are required to train it properly, leading to unacceptable overhead and hindering their applications especially in time varying environments.

Most recently, Kolmogorov-Arnold networks (KANs) have been proposed [18], based on the Kolmogorov-Arnold representation theorem, which decomposes high-dimensional functions into low-dimensional combinations through a unique structure, achieving efficient nonlinear function approximation. It motivates this work, aiming to build a more efficient network with smaller size for signal detection, enabling the training of the network with less pilot symbols but detection with higher performance. We will show that the use of a pure KAN is not efficient in dealing with the multi-user interference (MUI) in the MIMO system. Considering that the MUI is linear, we incorporate a linear NN layer to the KAN,

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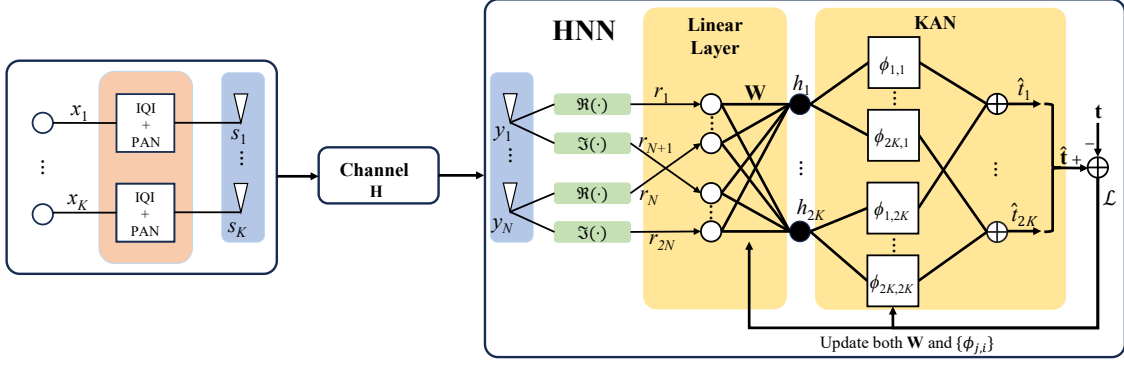


Fig. 1. Block diagram of the multi-user MIMO system with PAN and IQI, where the HNN is proposed for signal detection.

resulting in a hybrid NN-based signal detector, which handles the combined effects of MUI, IQI and PAN. The linear layer also significantly reduces the dimension of the signal, which is input to the KAN, leading to more efficient NN architecture. At the network training level, it is worth noting that traditional neural networks learn node weights, whereas KANs focus on learning activation functions [18]. The proposed HNN, which combines the linear layer of traditional NNs with KAN, can be trained within the backpropagation framework to achieve joint learning of both activation functions and linear weights. The derivations of the gradient backpropagation algorithm for the proposed HNN is presented in Section VI. Simulation results demonstrate that our HNN-based detector significantly outperforms existing detectors.

II. SYSTEM MODEL

Consider uplink transmission in a multi-user millimeter-wave MIMO system with K single-antenna user devices, each equipped with low cost modulators and PAs, that introduce PAN and IQI. At the receiver end (e.g., a base station (BS) or access point (AP)), N antennas are deployed

Define vector $\mathbf{x}(t) = [x_1(t), x_2(t), \dots, x_K(t)]^T$ as the signal symbols transmitted by K users at a given instant t , where $x_k(t)$ denotes the symbol transmitted by the k -th user at time t . The signal due to the impact of IQI can be represented as [17]

$$x_k^I(t) = \alpha_k x_k(t) + \beta_k x_k^*(t), \quad (1)$$

where

$$\alpha_k = \cos\left(\frac{a_k}{2}\right) + j b_k \sin\left(\frac{a_k}{2}\right) \quad (2)$$

$$\beta_k = b_k \cos\left(\frac{a_k}{2}\right) + j \sin\left(\frac{a_k}{2}\right), \quad (3)$$

a_k is the phase imbalance parameter, and b_k is the amplitude imbalance parameter. The imbalanced signal is then fed into the PA.

The nonlinear distortion of the power amplifier consists of two parts, i.e., $\psi(|x_k^I(t)|)$ and $\eta(|x_k^I(t)|)$ [19]. Among them, $\psi(|x_k^I(t)|)$ describes the amplitude-to-amplitude (AM/AM) conversion, i.e.,

$$\psi(|x_k^I(t)|) = \frac{\alpha_\psi |x_k^I(t)|}{(1 + (\alpha_\psi \frac{|x_k^I(t)|}{\lambda_\psi})^{2\sigma_\psi})^{\frac{1}{2\sigma_\psi}}}, \quad (4)$$

while $\eta(|x_k^I(t)|)$ describes the amplitude-to-phase (AM/PM) conversion, i.e.,

$$\eta(|x_k^I(t)|) = \frac{\alpha_\eta |x_k^I(t)|^{m_1}}{1 + (\frac{|x_k^I(t)|}{\lambda_\eta})^{m_2}}, \quad (5)$$

where α_ψ , λ_ψ , σ_ψ , α_η , λ_η , m_1 , and m_2 are parameters of the model. The distorted signal can then be expressed as

$$s_k(t) = \psi(|x_k^I(t)|) e^{j(\angle(x_k^I(t)) + \eta(|x_k^I(t)|))}, \quad (6)$$

where $\angle(x_k^I(t))$ denotes the phase of the complex signal $x_k^I(t)$. The received signal at time instant t is represented as

$$\mathbf{y}(t) = \mathbf{H}\mathbf{s}(t) + \mathbf{n}(t), \quad (7)$$

where $\mathbf{H} \in \mathbb{C}^{N \times K}$ is the MIMO channel matrix, $\mathbf{y}(t) = [y_1(t), y_2(t), \dots, y_N(t)]^T$, $\mathbf{s}(t) = [s_1(t), s_2(t), \dots, s_K(t)]^T$ and $\mathbf{n}(t)$ denotes a white Gaussian noise vector.

Since the signals used in the above model are complex-valued, we separate the real and imaginary parts of the signal and rearrange them into a real-valued form, i.e.,

$$\underbrace{\begin{bmatrix} \Re\{\mathbf{y}(t)\} \\ \Im\{\mathbf{y}(t)\} \end{bmatrix}}_{\mathbf{r}(t)} = \underbrace{\begin{bmatrix} \Re\{\mathbf{H}\} & -\Im\{\mathbf{H}\} \\ \Im\{\mathbf{H}\} & \Re\{\mathbf{H}\} \end{bmatrix}}_{\mathbf{H}'} \underbrace{\begin{bmatrix} \Re\{\mathbf{s}(t)\} \\ \Im\{\mathbf{s}(t)\} \end{bmatrix}}_{\mathbf{s}'(t)} + \underbrace{\begin{bmatrix} \Re\{\mathbf{w}(t)\} \\ \Im\{\mathbf{w}(t)\} \end{bmatrix}}_{\mathbf{n}'(t)}. \quad (8)$$

The input-output relationship of the MIMO system is complex due to the combined impacts of IQI, PAN and MUI which can be expressed as

$$\mathbf{r}(t) = \mathcal{S}(\mathbf{x}(t)) + \mathbf{n}'(t), \quad (9)$$

where $\mathcal{S}(\cdot)$ represents the system transfer function. The channel matrix, as well as the parameters of the IQI and PAN models, are assumed to be unknown. Each user transmits a pilot signal followed by data. The primary objective of the receiver at the BS/AP is to accurately detect the transmitted data symbols from all users.

III. PROPOSED HNN RECEIVER

We propose a HNN receiver based on KAN as shown in Fig. 1. Instead of using a pure KAN, we incorporate a conventional NN linear layer into KAN, which is more effective to handle the linear MUI introduced by the MIMO channel matrix. The linear layer also significantly reduces the dimension of the

signals as explained later. The linear layer is then followed by a KAN to handle the nonlinear distortions due to IQI and PAN. This leads to the HNN structure, which is used to learn the inverse function of $\mathcal{S}(\cdot)$ in (9) and jointly mitigate the MUI and nonlinear distortions. It is noted that the linear layer learns weight parameters directly, whereas the KAN module learns nonlinear activation functions. Despite this distinction, the overall HNN architecture remains fully trainable within the standard backpropagation framework.

The N -dimensional received signal is first processed by real and imaginary separation before being input into the HNN model. As shown in Fig. 1, the linear layer is used to deal with the linear MUI of the channel, yielding

$$\mathbf{h} = \mathbf{W}\mathbf{r}, \quad (10)$$

where $\mathbf{h} = [h_1, h_2, \dots, h_{2K}]^T$, $\mathbf{r} = [r_1, r_2, \dots, r_{2N}]^T$, and the weight matrix $\mathbf{W}$ has the size $2K \times 2N$. Considering the structure of $\mathbf{H}'$ in (8), we tie the entries of the weight matrix $\mathbf{W}$, so that we have

$$\mathbf{W} = \begin{bmatrix} \mathbf{W}^1 & \mathbf{W}^2 \\ -\mathbf{W}^2 & \mathbf{W}^1 \end{bmatrix}, \quad (11)$$

where $\mathbf{W}^1$ and $\mathbf{W}^2$ are matrices with dimension $K \times N$. The output of the linear layer contains $2K$ neurons, which also serve as the input of the subsequent KAN layer. Since K is typically much smaller than N in the system model, the linear layer also plays a role in data dimensionality reduction, reducing the dimensionality of the data input to the KAN layer. This significantly decreases the total number of model parameters, which will be demonstrated later.

After the signal is output from the linear layer, we assume that it has eliminated linear interference and contains only nonlinear distortions. Then, the distorted signal is input to the KAN layer, and the output from the j th node of KAN layer is represented as

$$\hat{t}_j = \sum_{i=1}^{2K} \phi_{j,i}(h_i), j = 1, 2, \dots, 2K, \quad (12)$$

where $\phi_{j,i}(\cdot)$ is the activation function connecting node h_i and $\hat{t}_j$. It is composed of a spline function and a basis function component $\mathbf{b}(h_i)$ that serves a role similar to a residual connection [18]. Specifically, the activation function is defined as

$$\phi_{j,i}(h_i) = \omega_s^{j,i} \text{spline}^{j,i}(h_i) + \omega_b^{j,i} \mathbf{b}(h_i), \quad (13)$$

where the basis function $\mathbf{b}(h_i)$ typically adopts the sigmoid linear unit (SiLU), defined as

$$\text{SiLU}(h_i) = \frac{h_i}{1 + e^{-h_i}}, \quad (14)$$

and the spline function $\text{spline}^{j,i}(h_i)$ is a weighted sum of multiple nonlinear k -th order B-spline basis functions $B_z(h_i)$ [18]:

$$\text{spline}^{j,i}(h_i) = \sum_{z=1}^Z c_z^{j,i} B_z(h_i), \quad (15)$$

where $Z = \text{grid} + k$, both grid and k are model parameters, and the weights ω_s , ω_b and c_z are trainable parameters. Hence,

the output signal of KAN layer can be written in the following form

$$\hat{\mathbf{t}} = \Phi(\mathbf{h}) = \begin{bmatrix} \sum_{i=1}^{2K} \phi_{1,i}(h_i) \\ \sum_{i=1}^{2K} \phi_{2,i}(h_i) \\ \vdots \\ \sum_{i=1}^{2K} \phi_{2K,i}(h_i) \end{bmatrix}, \quad (16)$$

where $\hat{\mathbf{t}} = [\hat{t}_1, \dots, \hat{t}_{2K}]^T$. Finally, by combining the estimated real and imaginary parts, the estimate of the transmitted signal can be reconstructed as $\hat{x}_k = \hat{t}_{2k-1} + j\hat{t}_{2k}$, $k = 1, 2, \dots, K$. The above design enables the HNN to transform the problem of fitting complex multivariate nonlinear functions into fitting multiple low-dimensional univariate functions. Through the superposition and combination of univariate functions, the proposed HNN can represent complex nonlinear functional relationships with higher precision, making it suitable for addressing combined effects of linear MUI due to the channel and nonlinear distortions induced by hardware impairments.

The training of the HNN can be implemented using back propagation. Suppose there are M_0 training samples $\{(\mathbf{r}(m), \mathbf{t}(m)), m = 1, 2, \dots, M_0\}$, where $\mathbf{t}(m) = [t_1(m), t_2(m), \dots, t_{2K}(m)]^T$ denotes the training symbols with $t_{2i-1} = \Re\{x_i\}$ and $t_{2i} = \Im\{x_i\}$, for $i = 1, 2, \dots, K$. The corresponding received signal is given by $\mathbf{r}(m) = [r_1(m), r_2(m), \dots, r_{2N}(m)]^T$. The loss function can be written as

$$\mathcal{L} = \frac{1}{M_0} \sum_{m=1}^{M_0} E(m), \quad (17)$$

where $E(m) = \frac{1}{2K} \sum_{i=1}^{2K} [t_i(m) - \hat{t}_i(m)]^2$. The loss function is calculated to update both the weight $\mathbf{W}$ and activation functions $\{\phi_{j,i}\}$ using back propagation.

IV. SIMULATION RESULTS

Assume that there are $K = 2$ users, each equipped with one antenna, and $N = 32$ antennas at the receiver end. Due to the limited space in this correspondence, we consider a flat-fading MIMO channel and note that the proposed method can be readily extended to frequency selective channels leveraging the orthogonal frequency division multiplexing technique. We use 16-quadrature amplitude modulation (QAM) in the simulations. The IQI and PAN are setup based on practical consideration, with the parameters detailed in [20]. The experiments are implemented using the deep learning framework PyTorch. A dataset consisting of 4000 samples is generated and it is split into training and testing sets in an 8:2 ratio. During training, we utilize the batch gradient descent strategy to optimize the model parameters with batch size 100. The adaptive moment estimation (Adam) optimizer is employed for parameter training. For the KAN layer of the proposed receiver, a single-hidden layer with 4 neurons is employed, with the model parameters setting grid = 5 and $k = 2$. Considering the characteristics of different models, we set different learning rates: for the HNN models, the learning rate is set to 0.0085, while for the ARVTDNN model, the learning rate is set to 0.005.

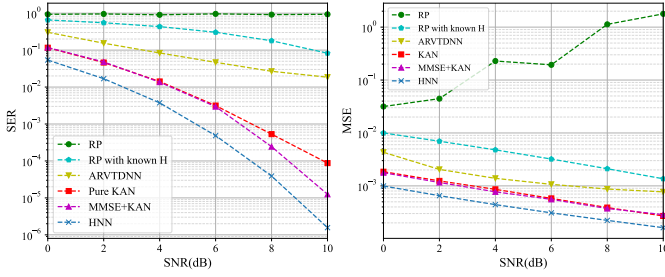


Fig. 2. SER and MSE performance comparisons of various receivers.

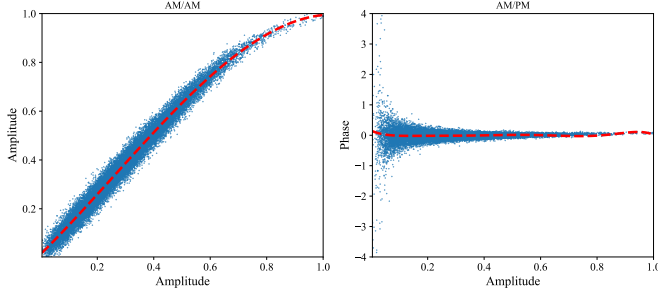


Fig. 3. AM/AM and AM/PM distortion characteristics curve of the collected dataset.

We evaluate the mean squared error (MSE) and symbol error rate (SER) performance of various methods under signal-to-noise ratio (SNR) conditions ranging from 0 to 10 dB. The evaluated methods include: (a) the proposed HNN; (b) pure KAN; (c) ARVTDNN [16]; (d) the minimum mean square error (MMSE)+KAN method (MMSE for channel equalization and KAN for dealing with nonlinearity); and (e) the real-valued polynomial (RP) method with a fifth order [21]. The left half of Fig. 2 illustrates the SER performance of these methods. It can be observed that the RP based method does not perform well due to the numerical instability. Even with a known channel matrix $\mathbf{H}$, the RP base method has poor SER performance compared to others. At the same time, the ARVTDNN has better performance, while it still has large gaps to the proposed HNN method. The MMSE+KAN method shows a modest performance improvement over pure KAN, suggesting that MMSE contributes to the mitigation of MUI. However, it still underperforms compared to the proposed HNN. This is because calculating MMSE weights requires a known nonlinearity, which is unavailable in practice, leading to a model mismatch issue that affects its performance. Meanwhile, the HNN method has over 4dB performance gain compared to the pure KAN, thanks to the introduction of the linear layer, which makes HNN more efficient in handling the combined effects of both linear MUI and nonlinear distortions. It is worth mentioning that the HNN method also has low complexity due to its small model size as shown in Table I. The right half of Fig. 2 shows the MSE performance of the aforementioned receivers. It can also be observed clearly that the proposed HNN method outperforms the state-of-the-art methods significantly.

To examine the performance under real scenario, we employ the collected dataset in [22], which includes I/Q signals from

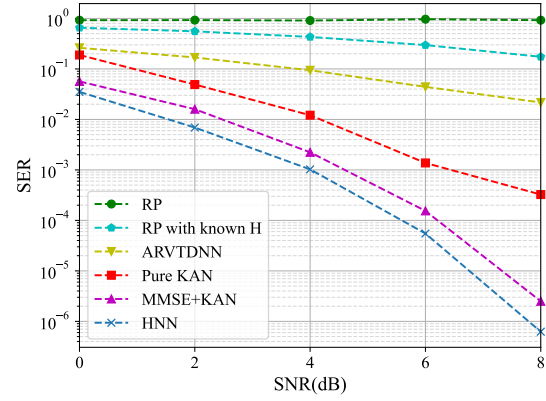


Fig. 4. SER performance comparisons of various receivers on the collected dataset.

TABLE I
PARAMETERS OF VARIOUS RECEIVERS.

Network	Nodes	Number of Parameters
Pure KAN	[64, 4]	3328
HNN	[64, 4, 4]	468
ARVTDNN	[480, 4, 4]	1944
MMSE+KAN	[4, 4]	464
RP	N. A.	2304

commercial PAs. Fig. 3 shows the AM/AM and AM/PM distortion characteristic curves from the dataset. The SER result is shown in Fig. 4. It can be seen that with the real data, the proposed HNN delivers outstanding performance compared to the state-of-the-art methods.

As the training and prediction complexities depend on the model size, Table I provides comparisons of the number of parameters used in various methods. It is evident that the number of parameters of the proposed HNN model is significantly lower than that of the pure KAN model and the RP method. The HNN has much fewer parameters than the ARVTDNN even with the same number of layers, because the input features of the ARVTDNN have a high dimension.

V. CONCLUSION

We have proposed a novel hybrid NN model, which combines a linear NN layer and a KAN layer, to handle the combined effects of the linear MUI and nonlinear hardware distortions in multi-user MIMO systems. We have shown that, compared to state-of-the-art methods, the proposed HNN model can have a much smaller size, while with significantly improved performance. Simulation results demonstrate the merits of the proposed HNN receiver.

VI. APPENDIX

Backpropagation is employed to train the proposed HNN, where gradient descent is executed not only on the linear weights at the edges, but also the trainable activation functions. Here we provide detailed derivations of the gradients in the hybrid network.

According to the chain rule, the partial derivative of the MSE E in (17) (we omit the time constant for simplicity)

with respect to the nonlinear activation function $\phi_{j,i}(h_u)$ and its input h_u can be expressed as

$$\frac{\partial E}{\partial \phi_{j,i}(h_u)} = \frac{\partial E}{\partial \hat{t}_v} \frac{\partial \hat{t}_v}{\partial \phi_{j,i}(h_u)}, \quad (18)$$

where $\frac{\partial E}{\partial \hat{t}_v} = 1/K$ and $\phi_{j,i}(h_u)$ is composed of trainable coefficients $\omega_s^{v,u}$, $\omega_b^{v,u}$ and $c_n^{v,u}$. In order to calculate the derivative to the nonlinear activation, we need to decompose the calculations with respect to these coefficients. Then, the partial derivative of E with respect to $\omega_s^{v,u}$ can be expressed as

$$\frac{\partial E}{\partial \omega_s^{v,u}} = \frac{\partial E}{\partial \hat{t}_v} \frac{\partial \hat{t}_v}{\partial \omega_s^{v,u}}, \quad (19)$$

where

$$\frac{\partial E}{\partial \hat{t}_v} = \frac{1}{K}, \quad \frac{\partial \hat{t}_v}{\partial \omega_s^{v,u}} = \text{spline}^{v,u}(h_u). \quad (20)$$

By combining equations (19) and (20), we obtain

$$\frac{\partial E}{\partial \omega_s^{v,u}} = \frac{1}{K} \text{spline}^{v,u}(h_u). \quad (21)$$

Similarly, we can obtain the partial derivative of E with respect to $\omega_b^{v,u}$,

$$\frac{\partial E}{\partial \omega_b^{v,u}} = \frac{1}{K} b(h_u). \quad (22)$$

The partial derivative of E with respect to $c_n^{v,u}$ can be expressed as

$$\frac{\partial E}{\partial c_n^{v,u}} = \frac{\partial E}{\partial \hat{t}_v} \frac{\partial \hat{t}_v}{\partial \text{spline}^{v,u}(h_u)} \frac{\partial \text{spline}^{v,u}(h_u)}{\partial c_n^{v,u}}, \quad (23)$$

where

$$\frac{\partial \hat{t}_v}{\partial \text{spline}^{v,u}(h_u)} = \omega_s^{v,u}, \quad \frac{\partial \text{spline}^{v,u}(h_u)}{\partial c_n^{v,u}} = B_n(h_u). \quad (24)$$

Thus, we have

$$\frac{\partial E}{\partial c_n^{v,u}} = \frac{1}{K} \omega_s^{v,u} B_n(h_u). \quad (25)$$

Then, we can obtain partial derivative of E with respect to the element $w_{u,v}$ in the u -th row and v -th column of $\mathbf{W}$ can be expressed as

$$\frac{\partial E}{\partial w_{u,v}} = \frac{1}{K} \sum_{p=1}^{2K} \frac{\partial \hat{t}_p}{\partial h_u} \frac{\partial h_u}{\partial w_{u,v}}, \quad (26)$$

where

$$\frac{\partial \hat{t}_p}{\partial h_u} = \omega_s^{p,u} \frac{\partial \text{spline}^{p,u}(h_u)}{\partial h_u} + \omega_b^{p,u} \frac{\partial b(h_u)}{\partial h_u}, \quad \frac{\partial h_u}{\partial w_{u,v}} = r_v \quad (27)$$

and

$$\frac{\partial \text{spline}^{p,u}(h_u)}{\partial h_u} = \sum_q c_q^{p,u} \frac{\partial B_q(h_u)}{\partial h_u}. \quad (28)$$

By combining equations (26), (27), and (28), we obtain

$$\begin{aligned} \frac{\partial E}{\partial w_{u,v}} = \frac{1}{K} \sum_{p=1}^{2K} \left\{ \omega_s^{p,u} \sum_q \left[c_q^{p,u} \frac{\partial B_q(h_u)}{\partial h_u} \right] r_v \right. \\ \left. + \omega_b^{p,u} \frac{\partial b(h_u)}{\partial h_u} r_v \right\} \end{aligned} \quad (29)$$

where $b(h_u)$ represents the SiLU function, and $B_q(h_u)$ denotes the B-spline function, which are both differentiable.

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Response to the Reviewers' Comments on

“Hybrid Neural Network-Based Signal Detection for Multi-User MIMO with Hardware Imperfections”

(Manuscript ID: VT-2025-02330)

Dawei Gao, Wenfeng Zhang, Qinghua Guo, Guisheng Liao, and Ming Jin

We first would like to thank the Editor for coordinating the review of the manuscript and the three Reviewers for the constructive comments. According to the comments, we have carefully revised the manuscript, which we believe has helped improve the quality of the manuscript significantly.

In the following, the reviewers' comments are addressed one by one. Comments by the reviewers are typed with italic fonts, while our responses are marked in blue. The revised parts of the manuscript are in red colour.

Reviewer 1's Comments:

ADDITIONAL COMMENTS TO THE AUTHOR:

The paper propose an hybrid neural network implementation for MU detection in the presence of transmitter impairments. The work use a concatenation of a linear NN to deal with the transformation of the propagation channel, then a KAN to deal with the nonlinear impairment.

In my opinion, the contribution of the paper is very limited and can be mainly summarised on an implemetation effort to combine the two networks.

Response: Thank you for your thoughtful comments. The primary contribution of this work is the introduction of a Kolmogorov-Arnold network (KAN) into signal detection to effectively address the nonlinear impairments introduced by low-cost transmitters. More importantly, we develop a hybrid neural network (HNN) that inspired by the signal flow to jointly mitigate multi-user interference (MUI) and nonlinear distortions. This design significantly reduces model complexity compared to a pure KAN, facilitating faster convergence with fewer pilot symbols. We hope the detailed responses below further clarify the contributions.

1. The training process of the proposed HNN is not clear. Ones the training is performed based on the signals degenerated by some transmitters, do we keep the same trained network to perform the decoding for other users with different transmitter impairments? Or do we need to train the network every time the users or the number of users changes?

Response: Thank you for your question. When the number or types of transmitters change, retraining of the HNN is required to adapt to the new nonlinear characteristics of the transmitter's impairments. However, the proposed HNN has substantially fewer trainable

parameters compared to other state-of-the-art models (as shown in Table I), making it highly suitable for online retraining with low overhead.

2. Same comments if the channel changes?

Response: In cases where only the channel changes but the number of users remains unchanged, it is sufficient to fine-tune only the linear layer of the HNN.

To validate this, we conducted an experiment with two different MIMO channels (channel matrices denotes as $\mathbf{H}_1$ and $\mathbf{H}_2$, respectively). Two datasets (denotes as D_1 and D_2) are generated using $\mathbf{H}_1$ and $\mathbf{H}_2$. Each dataset has the training length 3200 and the validation data length 800. We first train the proposed HNN using D_1 and evaluate its performance on both the D_1 and D_2 . Then, we freeze the KAN module of the trained HNN and fine tune the linear layer using the dataset D_2 . Finally, we evaluate the fine-tuned HNN on the D_2 . Fig. R1 illustrates the symbol error rate (SER) performance of various scenarios. The HNN without fine tuning performs well on the D_1 but, as expected, shows poor performance on the D_2 as the channel condition varies. In contrast, by freezing the KAN and retraining just the linear layer, we observe a significant performance gain on the new channel. This confirms the adaptability of the HNN with minimal retraining effort.

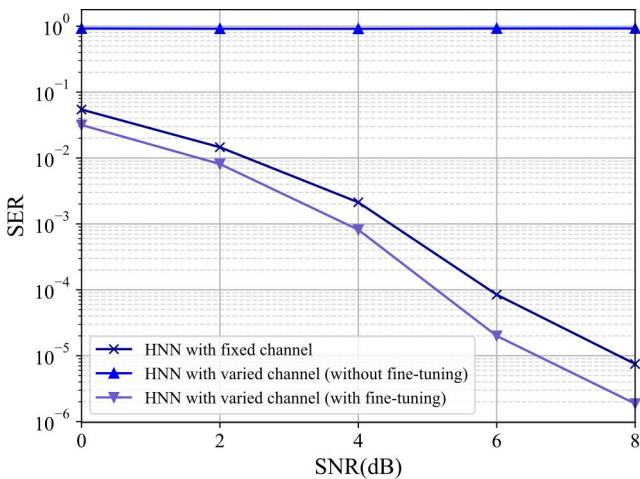


Fig. R1. SER performance of the proposed HNN with fine tuning on varied channel conditions.

3. In the Introduction, it is mentioned that the authors will should that the use of a pure KAN is not efficient in dealing with MUI. However, this point was only discussed through simulation without properly explaining it or proving it.

Response: The inefficiency of pure KAN in handling MUI is due to its high input dimensionality, which scales with the number of receive antennas. This leads to an excessive number of parameters, resulting in underfitting when limited pilots are available. The advantage of our HNN lies in using a linear layer to reduce dimensionality before applying KAN. This is empirically validated in Table I and Figs. 2 and 4.

4. From my understanding, the linear, NN plays the role of the receiver combiner. I wonder if it is possible to replace it with classical MMSE combiner then apply KAN to the output of the combiner. This would reduce the training and processing complexity.

Response: Thank you for your insightful suggestion. It is true that the linear layer works as a combiner for the linear MUI mitigation.

However, the proposed HNN is designed to simultaneously mitigate the MUI of MIMO channel and transmitter hardware impairments. To update the parameters of HNN, gradient descent and back-propagation are implemented on both the linear module and KAN module jointly.

The minimum mean square error (MMSE) is often used for linear equalization of MIMO channel with closed-form solution. If we replace the linear layer with MMSE combiner, the MMSE combiner and the KAN should be trained separately. In specific, the MMSE combiner matrix is calculated as

$$\mathbf{G} = \mathbf{R}_{rr}^{-1} \mathbf{R}_{rs},$$

where $\mathbf{R}_{rr} = E[\mathbf{r}\mathbf{r}^H]$ denotes the autocorrelation matrix of the received signal $\mathbf{r}$, and $\mathbf{R}_{rs} = E[\mathbf{r}\mathbf{s}^H]$ denotes the cross-correlation between the received signal $\mathbf{r}$ and the nonlinear distortion signal $\mathbf{s}$ at the transmitters.

However, the nonlinear distortion signal $\mathbf{s}$ at the transmitter is not available in practice. Instead, the original signal $\mathbf{x}$ is used as the target signal, but this leads to a severed nonlinear model mismatch issue. After the sub-optimal equalized signals are obtained, it is used as the input to the KAN for training. This is also cumbersome for the following training of KAN.

We have added the MMSE+KAN based method to the comparison of the proposed HNN and other state-of-the-art methods. The MSE and SER performance are shown in Fig. R2 and Fig. R3, respectively. It can be seen that the MMSE+KAN method has similar performance to the pure KAN while slightly fast convergence speed. But, the proposed HNN still provides outstanding performance gain compared to the MMSE+KAN, because the MMSE+KAN method suffers from model mismatch problems.

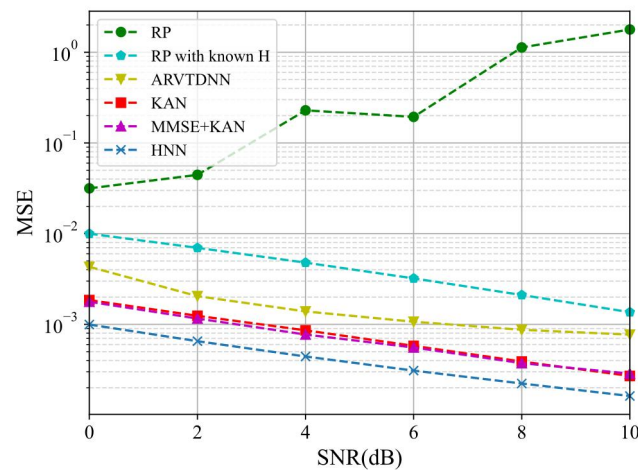


Fig. R2. MSE performance comparisons of various receivers.

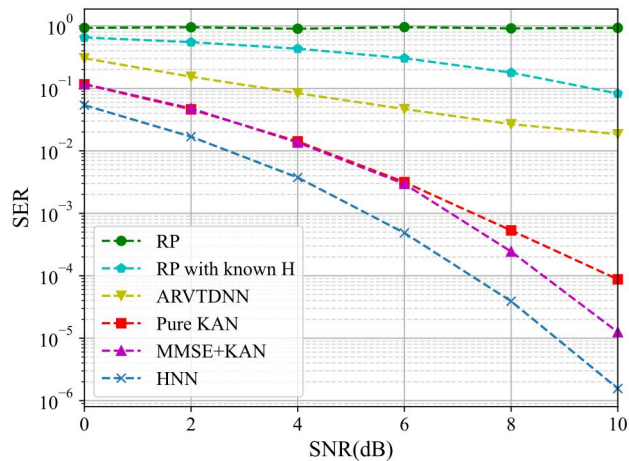


Fig. R3. SER performance comparisons of various receivers.

To examine the performance under real scenario, we employ the collected dataset in [R1]. The SER result of MMSE+KAN is evaluated and shown in Fig. R4. It can be seen that with the real data, the proposed HNN still delivers outstanding performance compared to the state-of-the-art methods.

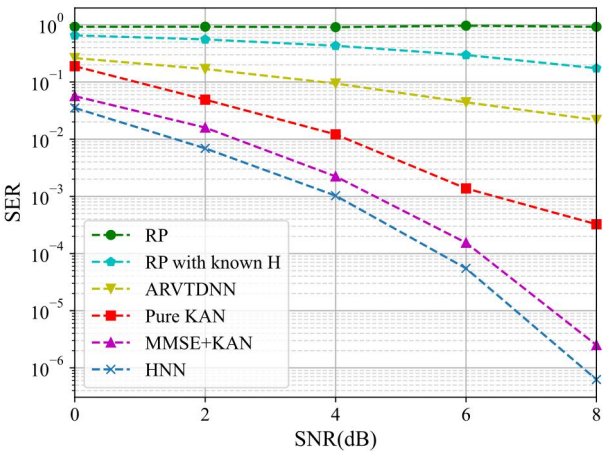


Fig. R4. SER performance comparisons of various receivers on the collected dataset.

According to your suggestion, we have incorporated the above results into the revised manuscript (Section IV, Page 3-4).

References:

[R1] Y. Wu, G. D. Singh, M. Beikmirza, L. C. N. de Vreede, M. Alavi, and C. Gao, "Opendpd: An open-source end-to-end learning & benchmarking framework for wideband power amplifier modeling and digital pre-distortion," in Proc. IEEE Int. Sym. Circ. Sys. (ISCAS), 2024, pp. 1 - 5.

5. Before equation (12), does it mean that the output of the linear layer is the transmitted signals from the users, $s(t)$, that already contain the nonlinear distortions of the transmitters.

Response: Yes, the linear layer aims to mitigate MUI, such that its output approximates the distorted transmitted signals $\mathbf{s}(t)$, which still contain the nonlinear effects of the hardware. The subsequent KAN module is then used to correct these distortions.

6. In the simulation results, the performance of the proposed HMM is compared with 3 other approaches. However, all the approaches are not relevant for the studied case, which makes the comparison not objective:

a. RP technique is developed to deal with PA nonlinearity, it is not appropriate for MU transmission which explain the obtained results

b. Pure KAN is good for nonlinear effects, but again it is not designed for MU detection.

I recommend combining RP with MMSE detection, or other appropriate method for this task.

Response: We appreciate your concern. Real-valued polynomial (RP) method has already been applied to signal detection in MIMO systems with nonlinear hardware impairments. Its core idea is to train RP coefficients individually for each transmit–receive antenna pair, enabling the mitigation of both nonlinear and linear distortions. For more details, please refer to Reference [R2] (also [21] in the manuscript).

Pure KAN is particularly advantageous for dealing with nonlinear problems. However, it is also capable of solving linear problems, since its activation functions are trainable and can be adapted to linearity. This property is illustrated in Figure 4.1 of Reference [R3] (also [18] in the manuscript).

According to your comments, we have evaluated the MMSE combiner with KAN detector in the response to Comment 4. The core issue is the conventional methods cannot be jointly designed and unavoidably meet model mismatch problem. Similarly, RP and MMSE cannot be jointly trained as well, because the nonlinear characteristics are often unknown.

Moreover, we fully understand the motivation for using MMSE to mitigate MUI and RP to handle nonlinear effects. Based on this idea, we included an evaluation of RP performance in the simulation under the condition where the channel matrix $\mathbf{H}$ is fully known. This setup represents the best-case scenario for RP+MMSE. However, as shown in Figs. R2, R3, and R4, even under such ideal conditions, RP still performs poorly due to its numerical instability issue.

References:

[R2] Z. A. Khan, E. Zenteno, P. Händel and M. Isaksson, “Digital predistortion for joint mitigation of I/Q imbalance and mimo power amplifier distortion,” IEEE Trans. Microw. Theory Techn., vol. 65, no. 1, pp. 322333, 2017.

[R3] Z. Liu, Y. Wang, S. Vaidya, F. Ruehle, J. Halverson, M. Soljačić, T. Y. Hou, and M. Tegmark, “Kan: Kolmogorov-arnold networks,” arXiv:2404.19756, 2024.

7. in the Appendix, after (19), the derivative of E with respect to t_{nu} is $1/2K$, and few lines later, the same derivative is $1/K$. Can you explain?

Response: Thank you for your careful review. This was a typo. The derivative of the E with respect to t_v should consistently be $1/K$, instead of $1/2K$. We have corrected this and

carefully proofread the entire appendix.

8. Before equation (7), the two numbers should be the same complex number.

Response: Thank you for the observation. We have corrected the superscript to ensure consistency and clarity (Section II, Page 2).

9. Before equation (14), definition of the basis function is not clear.

Response: According to your comment, we have improved the definition of basis function in the revised manuscript. The activation function now explicitly includes the spline function and a residual component based on a sigmoid linear unit (SiLU), as described in Section III of the revised manuscript.

10. In equation (17), is it a matrix $\times$ vector product? is it an element by element product?

Response: Thank you for your question. Equation (17) represents a nonlinear transformation, where each output node is a sum of nonlinear functions applied to inputs, i.e.,

$$\Phi(\mathbf{h}) = \begin{bmatrix} \phi_{1,1}(\cdot) & \phi_{1,2}(\cdot) & \cdots & \phi_{1,2K}(\cdot) \\ \phi_{2,1}(\cdot) & \phi_{2,2}(\cdot) & \cdots & \phi_{2,2K}(\cdot) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{2K,1}(\cdot) & \phi_{2K,2}(\cdot) & \cdots & \phi_{2K,2K}(\cdot) \end{bmatrix} \circ \mathbf{h}$$

$$= \begin{bmatrix} \sum_{i=1}^{2K} \phi_{1,i}(h_i) \\ \sum_{i=1}^{2K} \phi_{2,i}(h_i) \\ \vdots \\ \sum_{i=1}^{2K} \phi_{2K,i}(h_i) \end{bmatrix}$$

We have revised the text and the equation to better reflect this in the updated manuscript.

Reviewer 2's Comments

ADDITIONAL COMMENTS TO THE AUTHOR:

This paper proposes a signal detection method that integrates a neural network with the Kolmogorov – Arnold Network (KAN) to mitigate both inter-user interference and hardware impairments. Although the proposed approach is novel, the manuscript presents a limited contribution by focusing on a narrow set of scenarios, which raises concerns about the scalability and general applicability of the method.

Response: Thank you for your thoughtful comments and for acknowledging the novelty of our proposed approach. Regarding your concerns about scalability and generalizability, we have conducted extensive evaluations under varied conditions—including different channel realizations, numbers of users, and antennas—to demonstrate the robustness of our method. Details can be found in our responses to Comments 4 and 6. We also have updated the Section

IV of the revised manuscript to reflect these findings.

1. The manuscript does not provide a clear justification for addressing multi-user interference and I/Q imbalance separately. Could a unified neural network not be designed to jointly handle both impairments? In addition, the novelty of employing a linear layer in combination with KAN—rather than integrating a conventional linear receiver (e.g., Zero-Forcing) with KAN—remains insufficiently explained.

Response: Thank you for this important point. As shown in Fig. R5, the design of the HNN follows the actual signal flow: the transmitter-specific nonlinear distortions including IQI and PAN, followed by the MUI induced by the linear MIMO channel induced. Accordingly, we first apply a fully connected linear layer $\mathbf{W}$ to mitigate MUI, then use a KAN to mitigate IQI and PAN of users. This signal flow-driven architecture enables effective separation of tasks and improves interpretability. During training, the linear layers and KAN layers in the HNN are jointly trained to simultaneously using backpropagation.

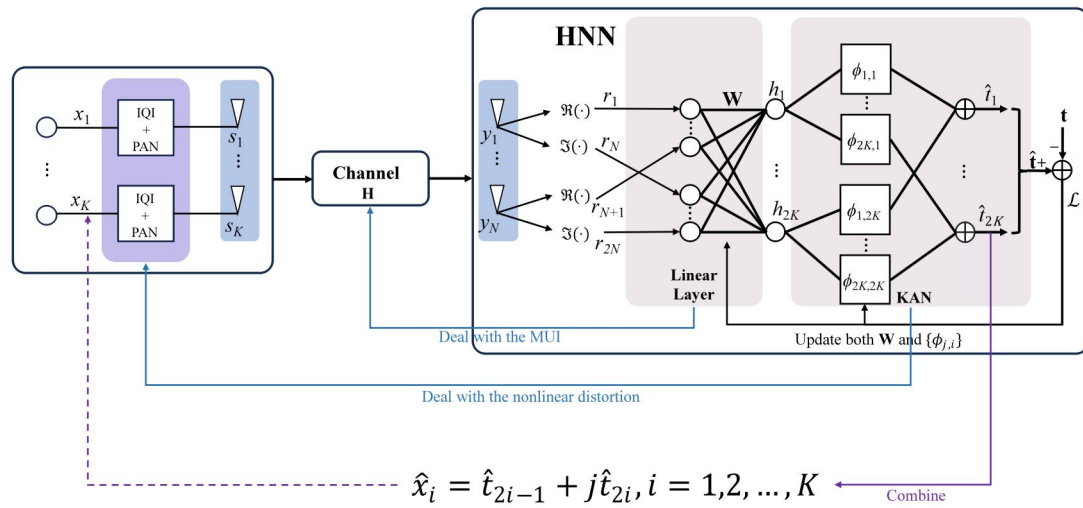


Fig. R5. Block diagram of the multi-user MIMO system with PAN and IQI, where the HNN is proposed for signal detection.

To validate the efficiency of proposed signal flow-based design, we compare the proposed HNN to the integration of a conventional linear receiver (i.e., MMSE) with KAN.

The MMSE is often used for linear equalization of MIMO channel with closed-form solution. If we replace the linear layer with MMSE combiner, the MMSE combiner and the KAN should be trained separately. In specific, the MMSE combiner matrix is calculated as

$$\mathbf{G} = \mathbf{R}_{rr}^{-1} \mathbf{R}_{rs},$$

where $\mathbf{R}_{rr} = \mathbb{E}[\mathbf{r}\mathbf{r}^H]$ denotes the autocorrelation matrix of the received signal $\mathbf{r}$, and $\mathbf{R}_{rs} = \mathbb{E}[\mathbf{r}\mathbf{s}^H]$ denotes the cross-correlation between the received signal $\mathbf{r}$ and the nonlinear distortion signal $\mathbf{s}$ at the transmitters.

However, in practice, the nonlinear distortion signal $\mathbf{s}$ at the transmitter is unknown. Instead, the original signal $\mathbf{x}$ is used as the target signal, but this leads to a severed nonlinear model

mismatch issue. After the sub-optimal equalized signals are obtained, it is used as the input to the KAN for training. This is also cumbersome for the following training of KAN. We have added the MMSE+KAN based method to the comparison of the proposed HNN and other state-of-the-art methods. The MSE and SER performance are shown in Fig. R6 and Fig. R7, respectively. It can be seen that the MMSE+KAN method has similar performance to the pure KAN while slightly fast convergence speed. But, the proposed HNN still provides outstanding performance gain compared to the MMSE+KAN, because the MMSE+KAN method suffers from model mismatch problems.

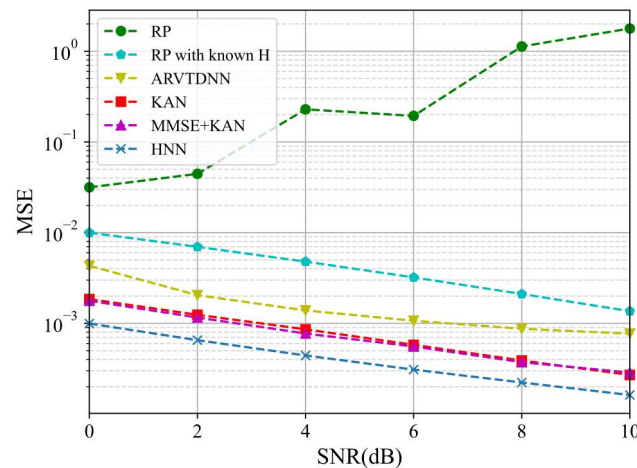


Fig. R6. MSE performance comparisons of various receivers.

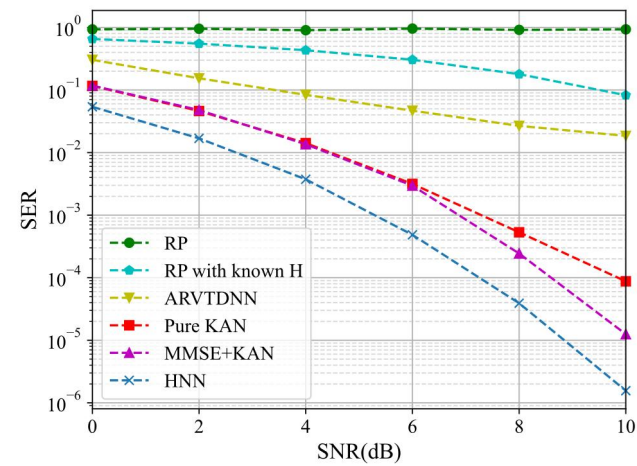


Fig. R7. SER performance comparisons of various receivers.

To examine the performance under real scenario, we employ the collected dataset in [R1]. The SER result of MMSE+KAN is evaluated and shown in Fig. R8. It can be seen that with the real data, the proposed HNN still delivers outstanding performance compared to the state-of-the-art methods.

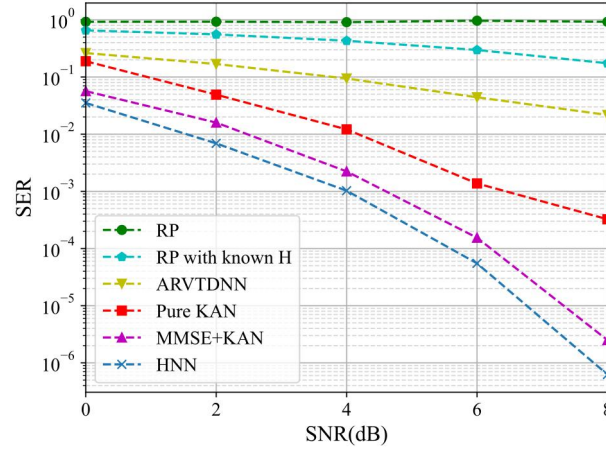


Fig. R8. SER performance comparisons of various receivers on the collected dataset.

According to your suggestion, we have incorporated the above results into the revised manuscript (Section IV, Page 3-4).

2. In Fig. 2, it is difficult to understand what's the detected signal per user; although there are K users, the number of detected signals is $2K$. Please clarify the detected signal per user.

Response: Thank you for your careful review and valuable comment. We consider K users transmitting complex signals and the aim of the proposed HNN is to detect these transmitted signals from these K users. However, most of the deep learning-based method and the commonly used training framework (e.g., Tensorflow or Pytorch) are more suited to real-valued computations. Therefore, we separate the real and imaginary parts of the received signal $\mathbf{r}$ as the input of the proposed HNN, resulting in an output dimension of $2K$, corresponding to the ones of K users. For constellation decision and SER evaluation, as shown in Fig. R5, we recombine the real and imaginary components of the estimated signals from the $2K$ -dimension output $\hat{\mathbf{t}} = [\hat{t}_1, \dots, \hat{t}_{2K}]^T$, i.e.,

$$\hat{x}_i = \hat{t}_{2i-1} + j\hat{t}_{2i}, i = 1, 2, \dots, K.$$

This clarification has been added to Section III of the revised manuscript.

3. Is the proposed scheme still valid in the mmWave system where hybrid beamforming would be the basic feature? The authors should mention the impact to the algorithm and performance by the hybrid beamforming in mmWave systems.

Response: We sincerely appreciate your insightful question. Yes, it is still valid. Although hybrid beamforming (HBF) is not currently incorporated into our model, the signal flow-driven architecture of the HNN offers strong flexibility for adaptation. As illustrated in Fig. R9, an analog combiner neural network as in [R4], denoted as $g_{\omega}(\cdot)$, can be introduced before the linear layer to realize analog beamforming. This analog combiner NN can be jointly trained with the HNN (which acts the digital part) using the backpropagation algorithm, where its parameters ω are updated accordingly. The extension, including hybrid beamforming used at both transmitter and receiver will be reported in another paper, due to the very limited space for this correspondence.

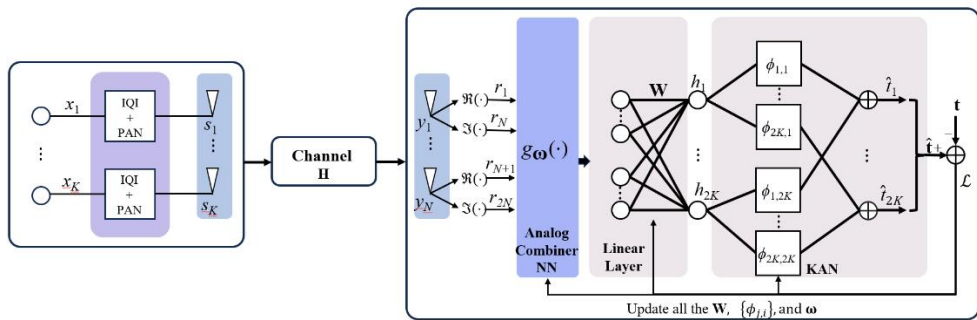


Fig. R9. Block diagram of the extended HNN architecture incorporating hybrid beamforming

References:

[R4] J. Chen, J. Tao, S. Luo, S. Li, C. Zhang, and W. Xiang, “A deep learning driven hybrid beamforming method for millimeter wave MIMO system,” Digit. Commun. Netw., vol. 9, no. 6, pp. 1291–1300, Dec. 2023.

4. In the multi-user MIMO, the set of users would be changed depending on the channel condition. If the user selection is included, should the model be retrained by the changed channel impulse response? Please describe the modification of the proposed scheme in the context of the user selection.

Response: Thank you for your comment. When the number of users changes, retraining is required. However, if the number of users remains fixed and only the channel conditions change, fine-tuning the linear layer is sufficient.

To validate this, we conducted an experiment with two different MIMO channels (channel matrices denotes as $\mathbf{H}_1$ and $\mathbf{H}_2$, respectively). Two datasets (denotes as D_1 and D_2) are generated using $\mathbf{H}_1$ and $\mathbf{H}_2$. Each dataset has the training length 3200 and the validation data length 800. We first train the proposed HNN using D_1 and evaluate its performance on both the D_1 and D_2 . Then, we freeze the KAN module of the trained HNN and fine tune the linear layer using the dataset D_2 . Finally, we evaluate the fine-tuned HNN on the D_2 . Fig. R10 illustrates the SER performance of various scenarios. The HNN without fine tuning performs well on the D_1 but, as expected, shows poor performance on the D_2 as the channel condition varies. In contrast, by freezing the KAN and retraining just the linear layer, we observe a significant performance gain on the new channel. This confirms the adaptability of the HNN with minimal retraining effort.

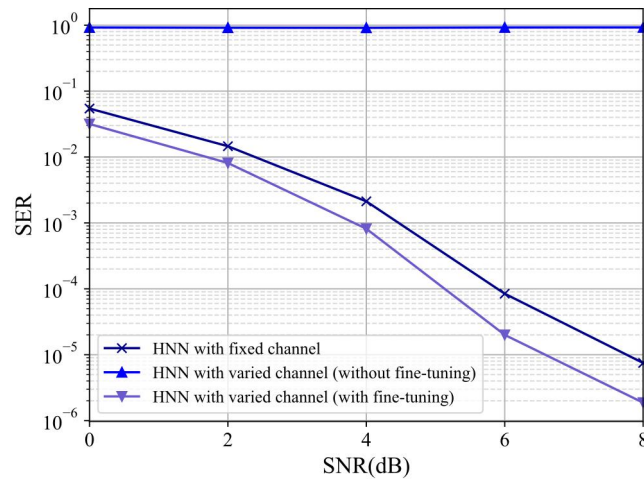


Fig. R10. SER performance of the proposed HNN with fine tuning on varied channel conditions.

5. If the proposed scheme does not require channel information, why do the users transmit pilots?

Response: Thank you for your constructive question. The proposed HNN does not rely on explicit channel estimation and it performs blind detection. However, pilot signals are still needed to train the neural network in a supervised manner. Once trained, the network can detect data symbols without needing channel state information.

6. Please evaluate the performance when the numbers of users and antennas become large.; is the proposed algorithm still valid with the different numbers of users and antennas?

Response: Thanks for your comments. In response to your comments, we have evaluated the performance of the HNN by varying the number of users K and the number of antennas N . Specifically, we simulate the SER performance of the HNN for $K = \{4, 6, 8\}$ and $\text{SNR} = \{2, 4, 6, 8\}$ with $N = \{32, 48\}$. The results are presented in Figs. R11 and R12. These figures demonstrate the HNN maintains strong performance even with increasing system scale, demonstrating the method's scalability and robustness.

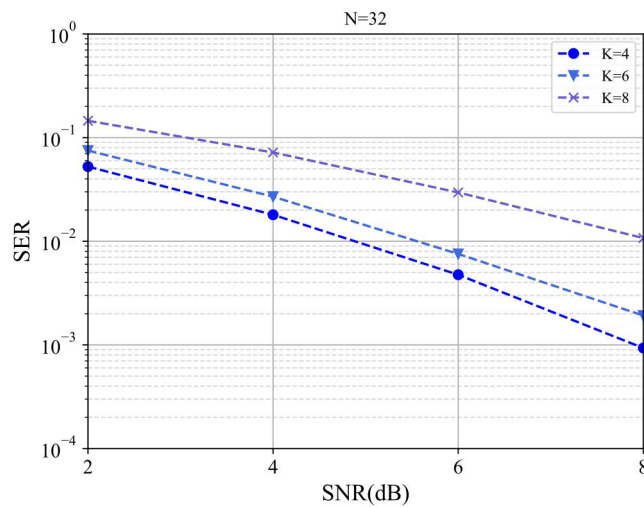


Fig. R11. SER performance of the HNN for $K = \{4, 6, 8\}$ with $N = 32$.

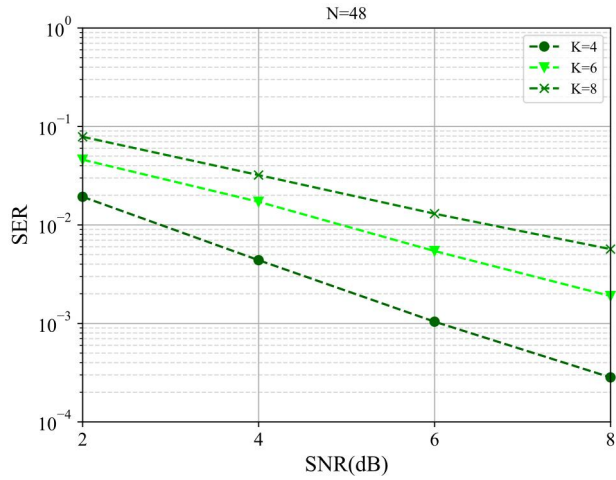


Fig. R12. SER performance of the HNN for $K = \{4, 6, 8\}$ with $N = 48$.

7. What’s the number of layers that brings the best performance in the context of the multi-user interference mitigation?

Response: Our validation experiments show that a single fully connected layer is sufficient to address MUI. Since the complexity of KAN increases with the number of layers, using fewer layers helps reduce the overall computational burden. Experimental results also demonstrate that a single-layer KAN is capable of effectively handling nonlinear distortions.

8. Numerous grammatical errors are present throughout the manuscript and should be carefully reviewed and corrected. For example,

- In page 3 of Section III, “It is note that...”
- In page 4 of Section IV, “we examine the mean square error (MSE) performance of various receivers is evaluated...”

Response: Thank you for your careful review and pointing this out. We have corrected the typographical errors and thoroughly proofread the entire manuscript.

Reviewer 3’s Comments:

ADDITIONAL COMMENTS TO THE AUTHOR:

Thank you for your effort!

Response: We sincerely appreciate your recognition of our work and your constructive feedback. We provide detailed responses to your specific concerns below.

1. In Section II, for the proposed system model and in Fig. 1, each user has a single antenna, whereas the paper aims to address a multi-user MIMO framework. The current system model is multi-user SIMO and not multi-user MIMO!

Response: Thank you for your comments. While each user has one antenna, the system as a whole, considering all users collectively, constitutes a multi-user MIMO system. This is a

common abstraction used in literature [R6], [R7], and [R8].

References:

[R6] G. Marti, T. Kölle, and C. Studer, “Mitigating smart jammers in multi-user mimo,” IEEE Trans. Signal Process., vol. 71, pp. 756 – 771, 2023.

[R7] M. Jiang and L. Hanzo, “Multiuser mimo-ofdm for next-generation wireless systems,” Proc. IEEE, vol. 95, no. 7, pp. 1430 – 1469, 2007.

[R8] E. Björnson, L. Sanguinetti, J. Hoydis, and M. Debbah, “Designing multi-user MIMO for energy efficiency: When is massive MIMO the answer?” in Proc. IEEE Wireless Commun. Netw. Conf. (WCNC), Apr. 2014, pp. 242–247.

2. *The mathematical modeling of hardware impairment (HWI) at transmitter is incomplete compared to realistic mmWave systems. Also, the receiver's HWIs, being inevitable in practice, have been taken for granted, giving rise to an insubstantial system model. It is worthwhile to point out that at mmWave and higher frequency bands, in addition to IQI and PAN, the transmitter suffers from phase noise (PN), and depending on the type of modulator, direct current offset (DCO) is another HWI at Tx in practical systems. The current transmission model has only considered IQI and PAN. Also, at the base station or access point, the received signals undergo IQI, DCO, PN, carrier frequency offset (CFO), and synchronization timing offset (STO). One of the state-of-the-art works in which all of the mentioned HWIs are jointly incorporated into the signal model at transceiver is given below. [1] A. D. Aref, C. Masouros, X. Zhu, E. Basar, S. Coleri, and H. Arslan, “Performance Analysis of OTSM under Hardware Impairments and Imperfect CSI,” IEEE Trans. Veh. Technol., vol. 73, no. 9, pp. 12991-13006, Sep. 2024.*

Response: Thank you for highlighting this. In our current work, we focused on modeling the dominant transmitter-side impairments, i.e., IQI and PAN, which are particularly pronounced in low-cost user devices. We agree that other impairments such as PN, CFO, and DCO are important, especially at mmWave frequencies. These are indeed worth investigating, and we plan to extend our framework to incorporate these impairments in future work. We have also cited [1] as suggested in the revised introduction (Section I).

3. *Reviewing the state-of-the-art should be improved, both in the context of mathematical modeling of HWIs and harnessing neural networks for detection in the presence of transceiver's HWIs. Reviewing the following paper and its references are recommended. [2] S. Liu, et al., “Hardware impairment estimation in NB-IoT: a parallel multi-task learning method,” IEEE IOT Journal, 2022.*

Response: We agree and appreciate the suggestion. We have carefully reviewed the additional recommended work [2] and included relevant discussions and citations in Section I of the revised manuscript to improve the literature review.

4. *The motivation for combining KAN with conventional linear neural network (NN) for signal detection under transceiver's HWIs should be articulated compared to the state-of-the-art hybrid NN models.*

Response: Thank you for the valuable suggestions. Based on your comments, we have added the MMSE+KAN based method to the comparison of the proposed HNN and other state-of-the-art methods. The MSE and SER performance are shown in Fig. R13 and Fig. R14, respectively. It can be seen that the MMSE+KAN method has similar performance to the pure KAN while slightly fast convergence speed. But, the proposed HNN still provides outstanding performance gain compared to the MMSE+KAN, because the MMSE+KAN method suffers from model mismatch problems.

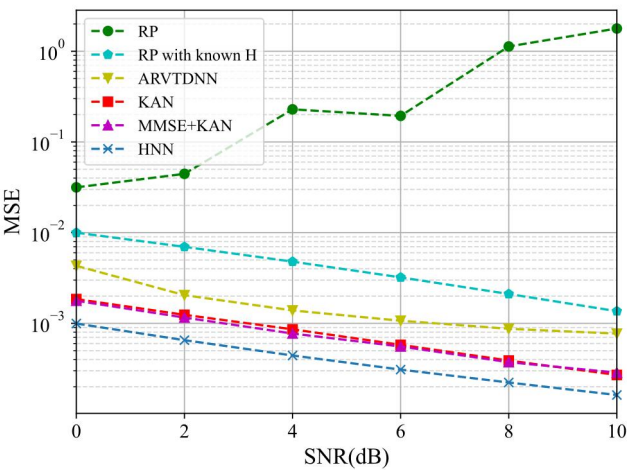


Fig. R13. MSE performance comparisons of various receivers.

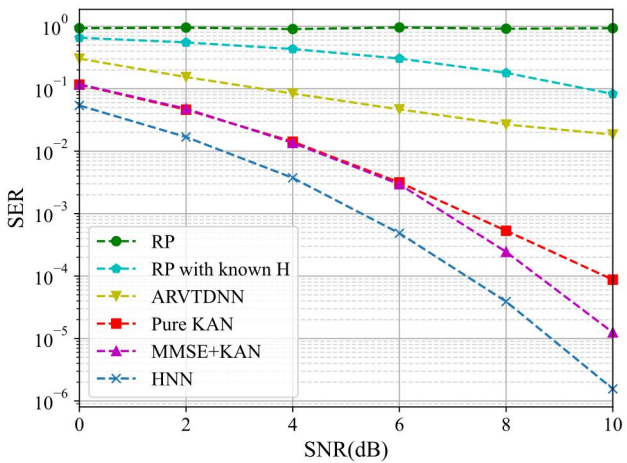


Fig. R14. SER performance comparisons of various receivers.

To examine the performance under real scenario, we employ the collected dataset in [R1]. The SER result of MMSE+KAN is evaluated and shown in Fig. R15. It can be seen that with the real data, the proposed HNN still delivers outstanding performance compared to the state-of-the-art methods.

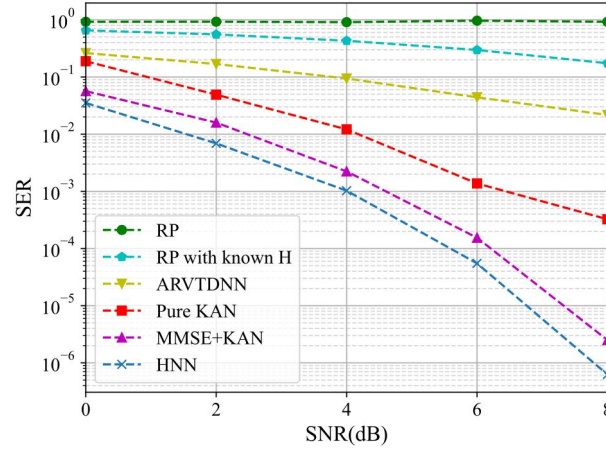


Fig. R15. SER performance comparisons of various receivers on the collected dataset.

According to your suggestion, we have incorporated the above results into the revised manuscript (Section IV, Page 3-4).

5. *The convergence of the proposed hybrid NN model should be verified under different scenarios.*

Response: We appreciate your valuable suggestions. In response to your comments, we have evaluated the performance of the HNN by varying the number of users K and the number of antennas N . Specifically, we simulate the SER performance of the HNN for $K = \{4, 6, 8\}$ and $\text{SNR} = \{2, 4, 6, 8\}$ with $N = \{32, 48\}$. The results are presented in Figs. R15 and R16. These figures demonstrate that the proposed HNN prolongs at different number of users and received antennas.

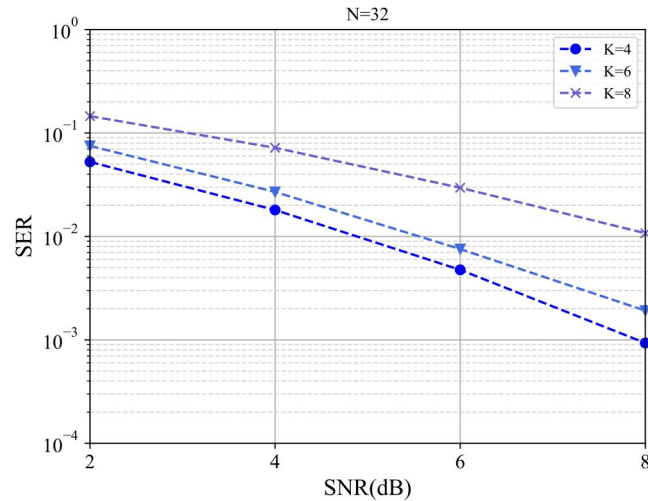


Fig. R15. SER performance of the HNN for $K = \{4, 6, 8\}$ with $N = 32$.

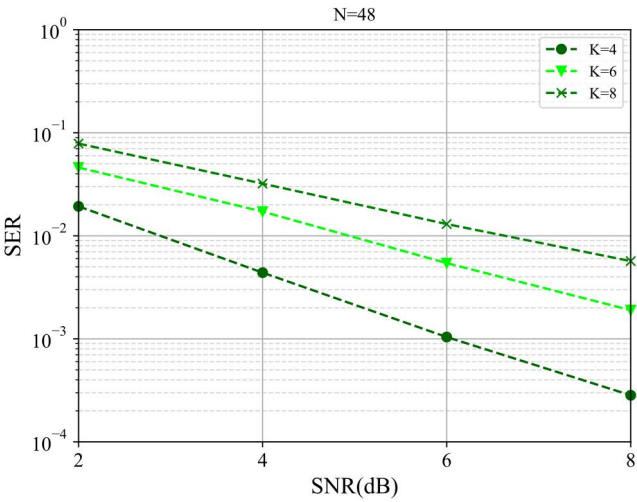


Fig. R16. SER performance of the HNN for $K = \{4, 6, 8\}$ with $N = 48$.

Hybrid Neural Network Based Signal Detection for Multi-User MIMO with Hardware Imperfections

Dawei Gao, Wenfeng Zhang, Qinghua Guo, Guisheng Liao, and Ming Jin

Abstract—This work deals with the issue of communication signal detection in multi-user multi-input multi-output (MIMO) with hardware imperfections, where low-cost users suffer from power amplifier nonlinearity (PAN) and in-phase and quadrature imbalance (IQI). To deal with the multi-user interference and nonlinear distortions due to hardware imperfections, we propose a novel signal detector using a hybrid neural network (HNN), which combines a Kolmogorov-Arnold network (KAN) and a conventional linear NN layer. The training of the HNN involves the update of both linear NN weights and nonlinear activation functions. It turns out that the proposed HNN detector with a much smaller model size significantly outperforms conventional polynomial-based techniques and state-of-the-art pure NN or KAN techniques.

Index Terms—Hardware imperfections, Kolmogorov-Arnold networks, multiple-input multiple-output, nonlinear distortion.

I. INTRODUCTION

As a core technology of modern wireless communication systems, multiple-input multiple-output (MIMO) significantly enhances channel capacity and transmission reliability by effectively utilizing the spatial dimension. However, as communication systems evolve toward higher frequency bands, larger bandwidths, and more complex modulation schemes, hardware impairments in low-cost radio frequency (RF) front-ends have become increasingly prominent, posing a critical bottleneck to further performance improvement in MIMO systems [1], [2], [3]. Hardware impairments in low-cost RF front-ends significantly degrade the performance of MIMO systems, especially in higher frequency bands and complex modulation schemes [4].

Among various hardware impairments, in-phase/quadrature (I/Q) imbalance is one of the most common types, manifesting as mismatches between the I and Q signal paths, thereby disrupting the orthogonality of the transmitted signals [5]. Meanwhile, the nonlinear characteristics of power amplifiers (PAs) leading to severe signal distortion cannot be overlooked, which is particularly pronounced in millimeter-wave and terahertz communication systems [3]. The combined effects of these hardware impairments further exacerbate signal distortion, posing significant challenges to the receiver design of

MIMO systems. To address the distortion issues caused by hardware impairments, various solutions have been proposed. For the nonlinear distortion caused by the PA, some classic polynomials-based methods have been investigated [6], [7]. These methods approximate the nonlinear characteristics of power amplifiers (PAs) by constructing polynomial models, and apply either predistortion at the transmitter to compensate for the nonlinear effects introduced by PAs, or postdistortion at the receiver to mitigate the distortion [8]. However, these approaches are prone to the numerical instability issue due to the matrix inversion in the determination of polynomial coefficients [9]. Other methods using the Wiener model [10] and the Hammerstein model [11] have also been studied, however, these approaches often result in model mismatch issues. Moreover, most existing works address I/Q imbalance (IQI) or PA nonlinearity (PAN) individually [12], [13], despite their concurrent occurrence in many cases, which necessitates handling their combined effects.

Methods based on artificial neural networks (NNs) have demonstrated significant potential in addressing nonlinear distortion in communication systems, thanks to their universal approximation capabilities. In [14], a parallel multi-task learning (MTL) estimator is developed to mitigate the coupling effects among different hardware impairments by utilizing their mutual correlations, which proves effective in compensating IQI. The work in [15] first introduced a real-valued focused time-delay neural network (RVFTDNN) to simulate PA's behavior, achieving promising performance. Building on this, extensive research [16], [17] has explored enhanced versions of RVFTDNN. Among these, the augmented RVFTDNN (ARVTDNN) model [16] improves the mitigation of PAN and IQI by incorporating higher-order signal components into the input of RVFTDNN. However, an issue with the ARVTDNN based techniques is that a large number of pilot symbols are required to train it properly, leading to unacceptable overhead and hindering their applications especially in time varying environments.

Most recently, Kolmogorov-Arnold networks (KANs) have been proposed [18], based on the Kolmogorov-Arnold representation theorem, which decomposes high-dimensional functions into low-dimensional combinations through a unique structure, achieving efficient nonlinear function approximation. It motivates this work, aiming to build a more efficient network with smaller size for signal detection, enabling the training of the network with less pilot symbols but detection with higher performance. We will show that the use of a pure KAN is not efficient in dealing with the multi-user interference (MUI) in the MIMO system. Considering that the MUI is linear, we incorporate a linear NN layer to the KAN,

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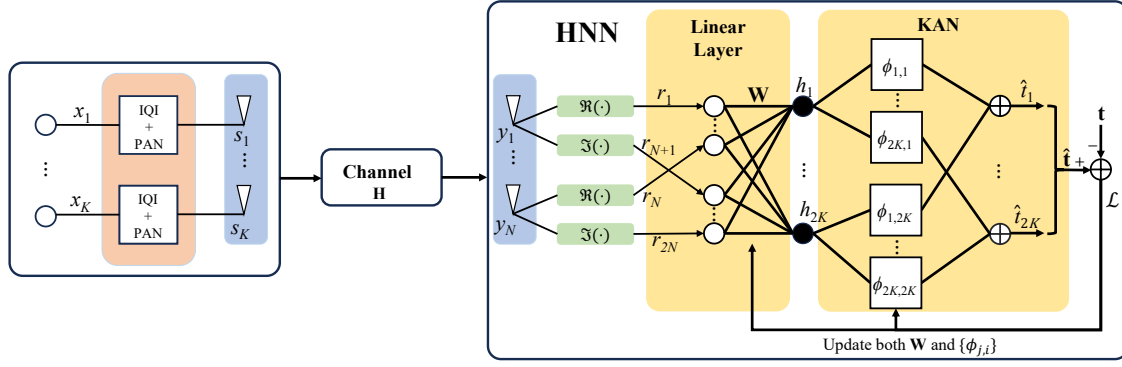


Fig. 1. Block diagram of the multi-user MIMO system with PAN and IQI, where the HNN is proposed for signal detection.

resulting in a hybrid NN-based signal detector, which handles the combined effects of MUI, IQI and PAN. The linear layer also significantly reduces the dimension of the signal, which is input to the KAN, leading to more efficient NN architecture. At the network training level, it is worth noting that traditional neural networks learn node weights, whereas KANs focus on learning activation functions [18]. The proposed HNN, which combines the linear layer of traditional NNs with KAN, can be trained within the backpropagation framework to achieve joint learning of both activation functions and linear weights. The derivations of the gradient backpropagation algorithm for the proposed HNN is presented in Section VI. Simulation results demonstrate that our HNN-based detector significantly outperforms existing detectors.

II. SYSTEM MODEL

Consider uplink transmission in a multi-user millimeter-wave MIMO system with K single-antenna user devices, each equipped with low cost modulators and PAs, that introduce PAN and IQI. At the receiver end (e.g., a base station (BS) or access point (AP)), N antennas are deployed

Define vector $\mathbf{x}(t) = [x_1(t), x_2(t), \dots, x_K(t)]^T$ as the signal symbols transmitted by K users at a given instant t , where $x_k(t)$ denotes the symbol transmitted by the k -th user at time t . The signal due to the impact of IQI can be represented as [17]

$$x_k^I(t) = \alpha_k x_k(t) + \beta_k x_k^*(t), \quad (1)$$

where

$$\alpha_k = \cos\left(\frac{a_k}{2}\right) + j b_k \sin\left(\frac{a_k}{2}\right) \quad (2)$$

$$\beta_k = b_k \cos\left(\frac{a_k}{2}\right) + j \sin\left(\frac{a_k}{2}\right), \quad (3)$$

a_k is the phase imbalance parameter, and b_k is the amplitude imbalance parameter. The imbalanced signal is then fed into the PA.

The nonlinear distortion of the power amplifier consists of two parts, i.e., $\psi(|x_k^I(t)|)$ and $\eta(|x_k^I(t)|)$ [19]. Among them, $\psi(|x_k^I(t)|)$ describes the amplitude-to-amplitude (AM/AM) conversion, i.e.,

$$\psi(|x_k^I(t)|) = \frac{\alpha_\psi |x_k^I(t)|}{(1 + (\alpha_\psi \frac{|x_k^I(t)|}{\lambda_\psi})^{2\sigma_\psi})^{\frac{1}{2\sigma_\psi}}}, \quad (4)$$

while $\eta(|x_k^I(t)|)$ describes the amplitude-to-phase (AM/PM) conversion, i.e.,

$$\eta(|x_k^I(t)|) = \frac{\alpha_\eta |x_k^I(t)|^{m_1}}{1 + (\frac{|x_k^I(t)|}{\lambda_\eta})^{m_2}}, \quad (5)$$

where α_ψ , λ_ψ , σ_ψ , α_η , λ_η , m_1 , and m_2 are parameters of the model. The distorted signal can then be expressed as

$$s_k(t) = \psi(|x_k^I(t)|) e^{j(\angle(x_k^I(t)) + \eta(|x_k^I(t)|))}, \quad (6)$$

where $\angle(x_k^I(t))$ denotes the phase of the complex signal $x_k^I(t)$. The received signal at time instant t is represented as

$$\mathbf{y}(t) = \mathbf{H}\mathbf{s}(t) + \mathbf{n}(t), \quad (7)$$

where $\mathbf{H} \in \mathbb{C}^{N \times K}$ is the MIMO channel matrix, $\mathbf{y}(t) = [y_1(t), y_2(t), \dots, y_N(t)]^T$, $\mathbf{s}(t) = [s_1(t), s_2(t), \dots, s_K(t)]^T$ and $\mathbf{n}(t)$ denotes a white Gaussian noise vector.

Since the signals used in the above model are complex-valued, we separate the real and imaginary parts of the signal and rearrange them into a real-valued form, i.e.,

$$\underbrace{\begin{bmatrix} \Re\{\mathbf{y}(t)\} \\ \Im\{\mathbf{y}(t)\} \end{bmatrix}}_{\mathbf{r}(t)} = \underbrace{\begin{bmatrix} \Re\{\mathbf{H}\} & -\Im\{\mathbf{H}\} \\ \Im\{\mathbf{H}\} & \Re\{\mathbf{H}\} \end{bmatrix}}_{\mathbf{H}'} \underbrace{\begin{bmatrix} \Re\{\mathbf{s}(t)\} \\ \Im\{\mathbf{s}(t)\} \end{bmatrix}}_{\mathbf{s}'(t)} + \underbrace{\begin{bmatrix} \Re\{\mathbf{w}(t)\} \\ \Im\{\mathbf{w}(t)\} \end{bmatrix}}_{\mathbf{n}'(t)}. \quad (8)$$

The input-output relationship of the MIMO system is complex due to the combined impacts of IQI, PAN and MUI which can be expressed as

$$\mathbf{r}(t) = \mathcal{S}(\mathbf{x}(t)) + \mathbf{n}'(t), \quad (9)$$

where $\mathcal{S}(\cdot)$ represents the system transfer function. The channel matrix, as well as the parameters of the IQI and PAN models, are assumed to be unknown. Each user transmits a pilot signal followed by data. The primary objective of the receiver at the BS/AP is to accurately detect the transmitted data symbols from all users.

III. PROPOSED HNN RECEIVER

We propose a HNN receiver based on KAN as shown in Fig. 1. Instead of using a pure KAN, we incorporate a conventional NN linear layer into KAN, which is more effective to handle the linear MUI introduced by the MIMO channel matrix. The linear layer also significantly reduces the dimension of the

signals as explained later. The linear layer is then followed by a KAN to handle the nonlinear distortions due to IQI and PAN. This leads to the HNN structure, which is used to learn the inverse function of $\mathcal{S}(\cdot)$ in (9) and jointly mitigate the MUI and nonlinear distortions. It is noted that the linear layer learns weight parameters directly, whereas the KAN module learns nonlinear activation functions. Despite this distinction, the overall HNN architecture remains fully trainable within the standard backpropagation framework.

The N -dimensional received signal is first processed by real and imaginary separation before being input into the HNN model. As shown in Fig. 1, the linear layer is used to deal with the linear MUI of the channel, yielding

$$\mathbf{h} = \mathbf{W}\mathbf{r}, \quad (10)$$

where $\mathbf{h} = [h_1, h_2, \dots, h_{2K}]^T$, $\mathbf{r} = [r_1, r_2, \dots, r_{2N}]^T$, and the weight matrix $\mathbf{W}$ has the size $2K \times 2N$. Considering the structure of $\mathbf{H}'$ in (8), we tie the entries of the weight matrix $\mathbf{W}$, so that we have

$$\mathbf{W} = \begin{bmatrix} \mathbf{W}^1 & \mathbf{W}^2 \\ -\mathbf{W}^2 & \mathbf{W}^1 \end{bmatrix}, \quad (11)$$

where $\mathbf{W}^1$ and $\mathbf{W}^2$ are matrices with dimension $K \times N$. The output of the linear layer contains $2K$ neurons, which also serve as the input of the subsequent KAN layer. Since K is typically much smaller than N in the system model, the linear layer also plays a role in data dimensionality reduction, reducing the dimensionality of the data input to the KAN layer. This significantly decreases the total number of model parameters, which will be demonstrated later.

After the signal is output from the linear layer, we assume that it has eliminated linear interference and contains only nonlinear distortions. Then, the distorted signal is input to the KAN layer, and the output from the j th node of KAN layer is represented as

$$\hat{t}_j = \sum_{i=1}^{2K} \phi_{j,i}(h_i), j = 1, 2, \dots, 2K, \quad (12)$$

where $\phi_{j,i}(\cdot)$ is the activation function connecting node h_i and $\hat{t}_j$. It is composed of a spline function and a basis function component $b(h_i)$ that serves a role similar to a residual connection [18]. Specifically, the activation function is defined as

$$\phi_{j,i}(h_i) = \omega_s^{j,i} \text{spline}^{j,i}(h_i) + \omega_b^{j,i} b(h_i), \quad (13)$$

where the basis function $b(h_i)$ typically adopts the sigmoid linear unit (SiLU), defined as

$$\text{SiLU}(h_i) = \frac{h_i}{1 + e^{-h_i}}, \quad (14)$$

and the spline function $\text{spline}^{j,i}(h_i)$ is a weighted sum of multiple nonlinear k -th order B-spline basis functions $B_z(h_i)$ [18]:

$$\text{spline}^{j,i}(h_i) = \sum_{z=1}^Z c_z^{j,i} B_z(h_i), \quad (15)$$

where $Z = \text{grid} + k$, both grid and k are model parameters, and the weights ω_s , ω_b and c_z are trainable parameters. Hence,

the output signal of KAN layer can be written in the following form

$$\hat{\mathbf{t}} = \Phi(\mathbf{h}) = \begin{bmatrix} \sum_{i=1}^{2K} \phi_{1,i}(h_i) \\ \sum_{i=1}^{2K} \phi_{2,i}(h_i) \\ \vdots \\ \sum_{i=1}^{2K} \phi_{2K,i}(h_i) \end{bmatrix}, \quad (16)$$

where $\hat{\mathbf{t}} = [\hat{t}_1, \dots, \hat{t}_{2K}]^T$. Finally, by combining the estimated real and imaginary parts, the estimate of the transmitted signal can be reconstructed as $\hat{x}_k = \hat{t}_{2k-1} + j\hat{t}_{2k}$, $k = 1, 2, \dots, K$. The above design enables the HNN to transform the problem of fitting complex multivariate nonlinear functions into fitting multiple low-dimensional univariate functions. Through the superposition and combination of univariate functions, the proposed HNN can represent complex nonlinear functional relationships with higher precision, making it suitable for addressing combined effects of linear MUI due to the channel and nonlinear distortions induced by hardware impairments.

The training of the HNN can be implemented using back propagation. Suppose there are M_0 training samples $\{(\mathbf{r}(m), \mathbf{t}(m)), m = 1, 2, \dots, M_0\}$, where $\mathbf{t}(m) = [t_1(m), t_2(m), \dots, t_{2K}(m)]^T$ denotes the training symbols with $t_{2i-1} = \Re\{x_i\}$ and $t_{2i} = \Im\{x_i\}$, for $i = 1, 2, \dots, K$. The corresponding received signal is given by $\mathbf{r}(m) = [r_1(m), r_2(m), \dots, r_{2N}(m)]^T$. The loss function can be written as

$$\mathcal{L} = \frac{1}{M_0} \sum_{m=1}^{M_0} E(m), \quad (17)$$

where $E(m) = \frac{1}{2K} \sum_{i=1}^{2K} [t_i(m) - \hat{t}_i(m)]^2$. The loss function is calculated to update both the weight $\mathbf{W}$ and activation functions $\{\phi_{j,i}\}$ using back propagation.

IV. SIMULATION RESULTS

Assume that there are $K = 2$ users, each equipped with one antenna, and $N = 32$ antennas at the receiver end. Due to the limited space in this correspondence, we consider a flat-fading MIMO channel and note that the proposed method can be readily extended to frequency selective channels leveraging the orthogonal frequency division multiplexing technique. We use 16-quadrature amplitude modulation (QAM) in the simulations. The IQI and PAN are setup based on practical consideration, with the parameters detailed in [20]. The experiments are implemented using the deep learning framework PyTorch. A dataset consisting of 4000 samples is generated and it is split into training and testing sets in an 8:2 ratio. During training, we utilize the batch gradient descent strategy to optimize the model parameters with batch size 100. The adaptive moment estimation (Adam) optimizer is employed for parameter training. For the KAN layer of the proposed receiver, a single-hidden layer with 4 neurons is employed, with the model parameters setting grid = 5 and $k = 2$. Considering the characteristics of different models, we set different learning rates: for the HNN models, the learning rate is set to 0.0085, while for the ARVTDNN model, the learning rate is set to 0.005.

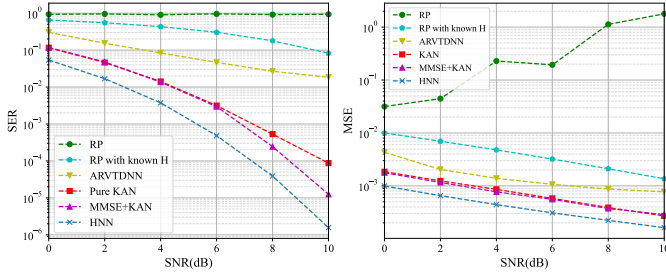


Fig. 2. SER and MSE performance comparisons of various receivers.

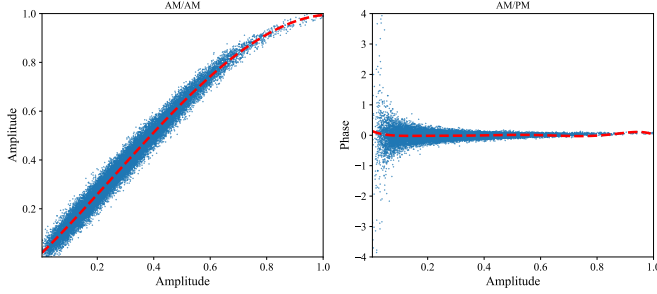


Fig. 3. AM/AM and AM/PM distortion characteristics curve of the collected dataset.

We evaluate the mean squared error (MSE) and symbol error rate (SER) performance of various methods under signal-to-noise ratio (SNR) conditions ranging from 0 to 10 dB. The evaluated methods include: (a) the proposed HNN; (b) pure KAN; (c) ARVTDNN [16]; (d) the minimum mean square error (MMSE)+KAN method (MMSE for channel equalization and KAN for dealing with nonlinearity); and (e) the real-valued polynomial (RP) method with a fifth order [21]. The left half of Fig. 2 illustrates the SER performance of these methods. It can be observed that the RP based method does not perform well due to the numerical instability. Even with a known channel matrix $\mathbf{H}$, the RP base method has poor SER performance compared to others. At the same time, the ARVTDNN has better performance, while it still has large gaps to the proposed HNN method. The MMSE+KAN method shows a modest performance improvement over pure KAN, suggesting that MMSE contributes to the mitigation of MUI. However, it still underperforms compared to the proposed HNN. This is because calculating MMSE weights requires a known nonlinearity, which is unavailable in practice, leading to a model mismatch issue that affects its performance. Meanwhile, the HNN method has over 4dB performance gain compared to the pure KAN, thanks to the introduction of the linear layer, which makes HNN more efficient in handling the combined effects of both linear MUI and nonlinear distortions. It is worth mentioning that the HNN method also has low complexity due to its small model size as shown in Table I. The right half of Fig. 2 shows the MSE performance of the aforementioned receivers. It can also be observed clearly that the proposed HNN method outperforms the state-of-the-art methods significantly.

To examine the performance under real scenario, we employ the collected dataset in [22], which includes I/Q signals from

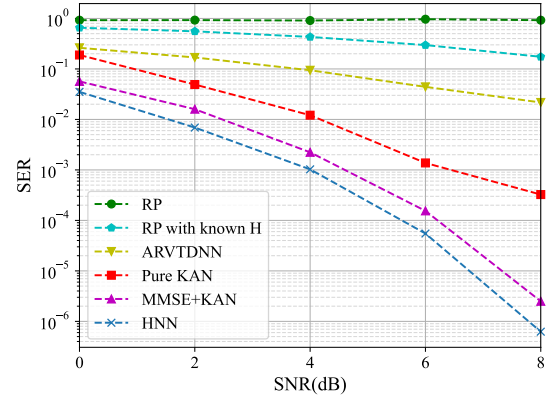


Fig. 4. SER performance comparisons of various receivers on the collected dataset.

TABLE I
PARAMETERS OF VARIOUS RECEIVERS.

Network	Nodes	Number of Parameters
Pure KAN	[64, 4]	3328
HNN	[64, 4, 4]	468
ARVTDNN	[480, 4, 4]	1944
MMSE+KAN	[4, 4]	464
RP	N. A.	2304

commercial PAs. Fig. 3 shows the AM/AM and AM/PM distortion characteristic curves from the dataset. The SER result is shown in Fig. 4. It can be seen that with the real data, the proposed HNN delivers outstanding performance compared to the state-of-the-art methods.

As the training and prediction complexities depend on the model size, Table I provides comparisons of the number of parameters used in various methods. It is evident that the number of parameters of the proposed HNN model is significantly lower than that of the pure KAN model and the RP method. The HNN has much fewer parameters than the ARVTDNN even with the same number of layers, because the input features of the ARVTDNN have a high dimension.

V. CONCLUSION

We have proposed a novel hybrid NN model, which combines a linear NN layer and a KAN layer, to handle the combined effects of the linear MUI and nonlinear hardware distortions in multi-user MIMO systems. We have shown that, compared to state-of-the-art methods, the proposed HNN model can have a much smaller size, while with significantly improved performance. Simulation results demonstrate the merits of the proposed HNN receiver.

VI. APPENDIX

Backpropagation is employed to train the proposed HNN, where gradient descent is executed not only on the linear weights at the edges, but also the trainable activation functions. Here we provide detailed derivations of the gradients in the hybrid network.

According to the chain rule, the partial derivative of the MSE E in (17) (we omit the time constant for simplicity)

with respect to the nonlinear activation function $\phi_{j,i}(h_u)$ and its input h_u can be expressed as

$$\frac{\partial E}{\partial \phi_{j,i}(h_u)} = \frac{\partial E}{\partial \hat{t}_v} \frac{\partial \hat{t}_v}{\partial \phi_{j,i}(h_u)}, \quad (18)$$

where $\frac{\partial E}{\partial \hat{t}_v} = 1/K$ and $\phi_{j,i}(h_u)$ is composed of trainable coefficients $\omega_s^{v,u}$, $\omega_b^{v,u}$ and $c_n^{v,u}$. In order to calculate the derivative to the nonlinear activation, we need to decompose the calculations with respect to these coefficients. Then, the partial derivative of E with respect to $\omega_s^{v,u}$ can be expressed as

$$\frac{\partial E}{\partial \omega_s^{v,u}} = \frac{\partial E}{\partial \hat{t}_v} \frac{\partial \hat{t}_v}{\partial \omega_s^{v,u}}, \quad (19)$$

where

$$\frac{\partial E}{\partial \hat{t}_v} = \frac{1}{K}, \quad \frac{\partial \hat{t}_v}{\partial \omega_s^{v,u}} = \text{spline}^{v,u}(h_u). \quad (20)$$

By combining equations (19) and (20), we obtain

$$\frac{\partial E}{\partial \omega_s^{v,u}} = \frac{1}{K} \text{spline}^{v,u}(h_u). \quad (21)$$

Similarly, we can obtain the partial derivative of E with respect to $\omega_b^{v,u}$,

$$\frac{\partial E}{\partial \omega_b^{v,u}} = \frac{1}{K} \mathbf{b}(h_u). \quad (22)$$

The partial derivative of E with respect to $c_n^{v,u}$ can be expressed as

$$\frac{\partial E}{\partial c_n^{v,u}} = \frac{\partial E}{\partial \hat{t}_v} \frac{\partial \hat{t}_v}{\partial \text{spline}^{v,u}(h_u)} \frac{\partial \text{spline}^{v,u}(h_u)}{\partial c_n^{v,u}}, \quad (23)$$

where

$$\frac{\partial \hat{t}_v}{\partial \text{spline}^{v,u}(h_u)} = \omega_s^{v,u}, \quad \frac{\partial \text{spline}^{v,u}(h_u)}{\partial c_n^{v,u}} = B_n(h_u). \quad (24)$$

Thus, we have

$$\frac{\partial E}{\partial c_n^{v,u}} = \frac{1}{K} \omega_s^{v,u} B_n(h_u). \quad (25)$$

Then, we can obtain partial derivative of E with respect to the element $w_{u,v}$ in the u -th row and v -th column of $\mathbf{W}$ can be expressed as

$$\frac{\partial E}{\partial w_{u,v}} = \frac{1}{K} \sum_{p=1}^{2K} \frac{\partial \hat{t}_p}{\partial h_u} \frac{\partial h_u}{\partial w_{u,v}}, \quad (26)$$

where

$$\frac{\partial \hat{t}_p}{\partial h_u} = \omega_s^{p,u} \frac{\partial \text{spline}^{p,u}(h_u)}{\partial h_u} + \omega_b^{p,u} \frac{\partial \mathbf{b}(h_u)}{\partial h_u}, \quad \frac{\partial h_u}{\partial w_{u,v}} = r_v \quad (27)$$

and

$$\frac{\partial \text{spline}^{p,u}(h_u)}{\partial h_u} = \sum_q c_q^{p,u} \frac{\partial B_q(h_u)}{\partial h_u}. \quad (28)$$

By combining equations (26), (27), and (28), we obtain

$$\begin{aligned} \frac{\partial E}{\partial w_{u,v}} = & \frac{1}{K} \sum_{p=1}^{2K} \left\{ \omega_s^{p,u} \sum_q \left[c_q^{p,u} \frac{\partial B_q(h_u)}{\partial h_u} \right] r_v \right. \\ & \left. + \omega_b^{p,u} \frac{\partial \mathbf{b}(h_u)}{\partial h_u} r_v \right\} \end{aligned} \quad (29)$$

where $\mathbf{b}(h_u)$ represents the SiLU function, and $B_q(h_u)$ denotes the B-spline function, which are both differentiable.

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